A Null-Representative Counterexample to the Proposed Young Convolution on $\mathbb{R}_I^\infty$

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Status. `candidate counterexample`; likely valid and ready for human review.

1 Source conjecture

Gill, Pantsulaia, and Zachary [1] define a sigma-finite measure λ_∞ on the set $\mathbb{R}_I^\infty = \mathbb{R}^\infty$ and state Young's inequality for

$$(f * g)(x) = \int_{\mathbb{R}_I^\infty} f(y)g(x-y)\, d\lambda_\infty(y).$$

After the proof of Theorem 5.10, on page 55 of the 65-page arXiv PDF, they write that the constant one is conjecturally sharp on $\mathbb{R}_I^\infty$.

> In closing we note that, Beckner [BE] and Brascamp-Lieb [BL] have shown that on $\mathbb{R}^n$ we can write Young's inequality as $\|f * g\|_r \leqslant (C_{p,q,r;n})^n \|f\|_p \|g\|_q$, where $C_{p,q,r;n} \leq 1$ is sharp. We conjecture that 1 is the sharp constant for $\mathbb{R}_I^\infty$.

The counterexample below shows a prior obstruction: the displayed convolution does not descend to the source's L^p spaces, because changing a function on a λ_∞-null set can change its convolution on a set of positive measure. Hence no sharp constant exists for the stated operation.

2 Definitions from the source

Set $I = [-1/2, 1/2]$, $I_n = \prod_{k>n} I$, and

$$D = \mathbb{R}_I^{\prime\infty} = \bigcup_{n\geq 1} (\mathbb{R}^n \times I_n), \qquad N = \mathbb{R}^\infty \setminus D.$$

Thus D consists of sequences whose coordinates lie in I eventually, and N consists of sequences having infinitely many coordinates outside I. The source equips N with the discrete topology, includes $\mathcal{P}(N)$ in its Borel sigma-algebra, and defines the measure of open sets by taking the supremum of finite-dimensional compact pieces contained in them. Consequently

$$\lambda_\infty(N) = 0. \tag{1}$$

Indeed every positive compact piece used in the definition lies in D, so none is contained in the disjoint open set N.

Let $I_0 = I^{\mathbb{N}}$. The normalization in the source is $\lambda_\infty(I_0) = 1$, and the coordinate marginals of $\lambda_\infty|_{I_0}$ are normalized Lebesgue measure on I. Write $\mu = \lambda_\infty|_{I_0}$.

3 Proof intuition

The measure is concentrated on sequences that eventually remain in the tail box I. Subtraction destroys that property: for two independent tail-box coordinates, the difference exits I with probability 1/4. Across infinitely many independent coordinates, exits occur infinitely often almost surely. Therefore the difference of two typical points of I_0 lies in the null complement N.

This turns the null-set indicator $\mathbf{1}_N$ into a decisive witness. It is the zero element of every L^q, but the pointwise convolution formula samples it at typical differences and returns one on I_0.

4 Formal counterexample

Lemma 1. *For $\mu \otimes \mu$-almost every $(x, y) \in I_0 \times I_0$, one has $x - y \in N$. Equivalently,*

$$(\mu \otimes \mu)\{(x, y) : x - y \in D\} = 0.$$

Proof. For $k \geq 1$ let

$$B_k = \{(x, y) \in I_0^2 : |x_k - y_k| > 1/2\}.$$

The sets B_k depend on disjoint coordinate pairs and hence are independent. The two triangles outside the diagonal strip of width 1/2 in the unit square $I \times I$ have total area 1/4, so $(\mu \otimes \mu)(B_k) = 1/4$.

The condition $x - y \in D$ says that only finitely many B_k occur. For every m,

$$(\mu \otimes \mu)\left(\bigcap_{k>m} B_k^c\right) = \lim_{M\to\infty} \left(\frac{3}{4}\right)^{M-m} = 0.$$

Taking the countable union over m proves the claim. □

Theorem 1. *The convolution formula in Theorems 5.10 and 5.13 of [1] is not well-defined on almost-everywhere equivalence classes. More precisely, for every admissible exponent triple $p, q, r \in [1, \infty]$, there are measurable representatives f and g such that*

$$\|f\|_p = 1, \qquad \|g\|_q = 0, \qquad \|f * g\|_r \geq 1.$$

Consequently the claimed Young inequality, and hence the conjecture that its sharp constant is one, is false or ill-posed as stated.

Proof. Take

$$f = \mathbf{1}_{I_0}, \qquad g = \mathbf{1}_N.$$

Both functions are measurable under the source's sigma-algebra. The normalization $\lambda_\infty(I_0) = 1$ gives $\|f\|_p = 1$ for every p, while (1) gives $\|g\|_q = 0$ for every q, including $q = \infty$ in the essential-supremum sense.

Lemma 1 and Fubini give, for μ-almost every $x \in I_0$,

$$\mu\{y \in I_0 : x - y \in N\} = 1.$$

For those x the source's pointwise formula yields

$$(f * g)(x) = \int_{\mathbb{R}_I^\infty} \mathbf{1}_{I_0}(y)\mathbf{1}_N(x-y)\, d\lambda_\infty(y) = 1.$$

Thus $\|f * g\|_r \geq 1$ for every r. But g and the identically zero function are equal λ_∞-almost everywhere, whereas convolution with the zero representative is identically zero. The formula therefore does not define an operation on L^p equivalence classes, and its asserted norm bound has right-hand side zero and left-hand side at least one. $\square$

5 Where the source proof fails

The proof of Theorem 5.13 changes variables as though

$$\int g(x-y)\, d\lambda_\infty(x) = \int g(x)\, d\lambda_\infty(x)$$

for the relevant y. However, the source itself identifies ℓ_1 as the maximal group of admissible translations for λ_∞. A typical tail-box point is not in ℓ_1, and Lemma 1 quantifies the resulting singular behavior. Fubini alone cannot supply the missing translation invariance.

6 Verification, limitations, and novelty

The argument is exact and contains no numerical or unproved dependency. The finite-coordinate check is the elementary area computation $\Pr(|X - Y| > 1/2) = 1/4$ for independent uniform $X, Y \in I$; countable additivity then gives Lemma 1.

The result attacks the convolution and Young inequality exactly as written. It does not preclude a different construction using a restricted translation group, a different measure, or an explicit representative convention. Such a change would be a reformulation, not a repair of the stated L^p operation.

For novelty, on 19 July 2026 we searched the run's cheap indexes for the arXiv id and title, then searched the web for the exact title with the phrases `convolution`, `Young`, `sharp constant`, `critique`, and `R_Iînfty`. The search found the source and citing works but no correction or prior counterexample. This was a bounded, not exhaustive, search.

Human-review recommendation. Review the source's null-complement measure assignment and whether any unwritten canonical-representative convention was intended. Under the published definitions the counterexample is complete; an added convention would instead expose that the source is defining a different, highly degenerate operation.

References

[1] T. L. Gill, G. R. Pantsulaia, and W. W. Zachary, *Constructive Analysis in Infinitely many variables*, arXiv:1206.1764 (2012), especially Sections 2 and 5, Theorems 5.10 and 5.13, and PDF page 55.